

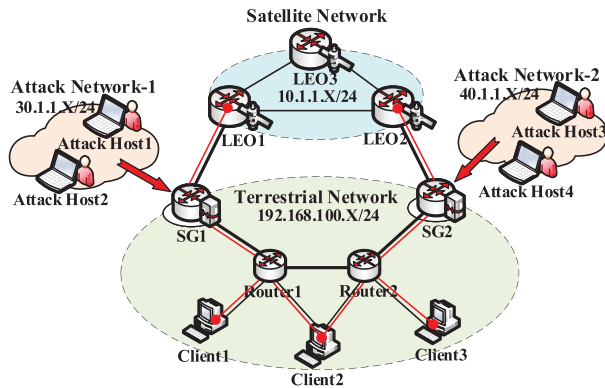


FIGURE 1. Testbed architecture.

node and seven computing nodes. The servers we use are DELL PowerEdge R720 rack-mount servers with Intel Xeon E5-2609 CPU, 32G memory and 1 TB hard-disk storage. Linux bridge is used to implement routing and switching between hosts. Next, we will introduce the testbed in three parts: satellite network, terrestrial network, and attack network.

### 1) SATELLITE NETWORK

Because this paper focuses on the identification of malicious traffic at the satellite network entrance, the scale of the satellite network is simplified in the testbed. We only use three Low Earth Orbit (LEO) satellites in the same orbit as test nodes. LEO1 and LEO2 remain connected to the satellite gateway1 (SG1) and SG2 of the terrestrial network, and the LEO satellites are always visible.

To make our testbed a real scenario, we simulate the actual latency and high-latency transmission protocols of the satellite network. We implement DTN in the satellite network to meet the need for high transmission delay with the help of Interplanetary Overlay Network (ION) [33], [34]. Then, we limit the transmission delay between LEO to 50ms and the transmission delay between LEO and SG to 3ms with the help of the Linux Traffic Control.

### 2) TERRESTRIAL NETWORK

As is shown in Fig. 1, the terrestrial network contains two SGs, two routers, and three clients. When deploying the terrestrial backbone network, considering that this paper focuses on designing security data sets based on traffic features, a simple star topology is used to avoid packet loss.

We use the TCP/IP protocol stack in the terrestrial network because it is used in the terrestrial networks in most cases. To facilitate protocol conversion between the satellite and terrestrial network, we develop a DTN-TCP/IP bidirectional conversion and deploy this function on two SGs. Moreover, the Tcpdump tool is installed on two SGs to capture the pcap files of the simulation uptime. Three clients are used to communicate with the satellite network to form benign traffic during the test time.

### 3) ATTACK NETWORK

In order to truly represent the modern threat environment, we installed different operating systems (Ubuntu14.04, Ubuntu16.04, Ubuntu18.04, and Win10) on the four attack hosts and set them in different network segments.

Since our proposed data set aims to test the attack detection technology of the STIN, it should cover a diverse set of network attack techniques and scenarios.

The terrestrial network has many security problems, and a DDoS attack is undoubtedly the most serious harm to it. Therefore, we hope that the malicious attacks on the terrestrial network can cover as many types as possible. We generated a total of nine different attack scenarios based on open source software simulations, including Botnets, Web Attacks, Backdoor, and six different DDoS attacks (LDAP, MSSQL, NetBIOS, Portmap, Syn, UDP).

Unlike terrestrial networks, satellite networks have a higher level of security and unique DTN transmission protocol, so malicious attacks applicable to terrestrial networks are difficult to enter into satellite networks and cause damage. However, DDoS attacks are flexible and changeable. Only by exploiting the loopholes in the DTN protocol's forwarding mechanism, it can greatly affect satellite communications. Since our prototype was implemented based on Interplanetary Overlay Network (ION) [33] software, we designed and implemented a DDoS attack for the DTN protocol, which is a cancellation transmission mechanism based on the LTP protocol. There is a loophole in the design of ION's LTP protocol, that is, after any party receiving and receiving receives Cancel\_segment, it must return Cancel\_ACK\_segment in the direction of the connecting node that is communicating, without additional conditions. As soon as the node receives Cancel\_segment, it must return a confirmation message. As shown in Fig. 2(a), the satellite network has a high resource occupancy rate during communication under normal conditions. When the attacking node launches an attack on the satellite network, as shown in Fig. 2(b), the attacker's control host will send a large number of Cancel\_segments to the SG, it will force ION to reply to Cancel\_ACK\_segment to the party that normally communicates, occupying the bandwidth and computing resources of the link, thereby implementing the DDoS attack for the satellite network. Because the DTN transmission mechanism is more complex, only two types of satellite network DDoS attacks based on Syn and UDP are constructed.

## B. SECURITY DATA SET DESIGN

Our proposed STIN security data set aims to collect various types of attacks in modern satellite and terrestrial network environments. We construct the data set based on the test architecture in Fig. 1. Meanwhile, to fully restore the modern threat environment, we set the experiment duration to 3 hours, starting at 15:00 and ending at 18:00. During the experiment, we kept each domain Transmission of benign traffic. As shown in Table 1, during the experiment, we performed